Infectious Keratitis: Literature Review and Recent Trends in Management

Bulus SS,[1*] Sarki PD,[2] Mahmoud Z[2]

ABSTRACT

Infectious Keratitis is responsible for over two million cases of blindness, annually, worldwide. Though rare in developed countries, it is a major problem in developing countries of Africa and the Indian sub-continent with most of the burden attributed to trauma, low social-economic status, illiteracy and contact lens wear among others. Its effect differs from continent to continent being higher in Africa and India. Causative agents include bacteria, such as staphylococcus, virus such as herpes simplex, fungus such as aspergilus and parasites such as acanthamoeba. Diagnosis may be clinical but also involves culture on various media such as blood agar, in-vivo confocal microscopy for fungal and acanthamoeba agents, polymerase chain reaction for sensitivity in detecting infective agents. Anterior segment optical coherence tomography to provide information on corneal infiltrates and scar size. Fourth generation fluoroquinolones such as gatiflox are becoming more useful as single agents for treatment. Bandage contact lens is also a useful approach for enhancing wound healing. Newer trends involving use of ultraviolet radiation prior to drug application, combination of fluoroquinolones with Amphotericin B along with drug eluting mechanism along with micro-needle imprint are also becoming useful in treatment. Newer pharmaceuticals such as nano-particles, antimicrobial peptides (AMB) thymosin are useful along with newer systemic antibiotics combinations. With these approaches, there is hope in the treatment of infectious keratitis.

Keywords: Infectious Keratitis, Novel approach to management, Literature review on recent trends

INTRODUCTION

Keratitis is the inflammation of the cornea characterized by corneal oedema, infiltration of inflammatory cells and ciliary congestion.[1]

Infectious Keratitis (IK) is the inflammation of the cornea as a result of infectious agents like; Bacteria such as staphylococcus specie (spp), streptococcus spp, pseudomonas or viruses such as herpes simplex, varicella zoster or fungal agents like candida spp, aspergillus spp and fusarium spp. It may also be caused by protozoa such as acanthamoeba spp.

It is responsible for more than 2 million cases of blindness annually around the world with associated corneal ulcer, stromal infiltrates and hypopyon.

Department of Ophthalmology,[1] Faculty of Clinical Sciences, College of Medicine, Kaduna State University. Department of Ophthalmology,[2] Barrau Dikko Teaching Hospital, Kaduna State University.

*Corresponding Author: *stanley.bulus@kasu.edu.ng*

Date manuscript was received: 14/5/25
Date manuscript was accepted: 10/7/25

IK though rare in developed countries is a major problem in developing countries of Africa and Indian sub-continent.[2]

IK has been shown to be the commonest cause of blindness world-wide with most of the burden attributed to increased risk of trauma, low socio-economic status and illiteracy.[3]

INCIDENCE

Incidence of IK has been reported to be in the range of 2.5 – 27.6 per 100,000 population/year in the U.S.[4] and 2.6 – 40.3 per 100,000 population/year in the U.K.[5] Higher incidences of 113 per 100,000 and 799 per 100,000 population were reported in Southern India and Nepal respectively.[6,7] which is similar to Africa due to poor environment and personal hygiene, lower educational level and increased risk at work related corneal trauma.

The table below summarizes the reported annual incidence of IK per 100,000 population across various countries of the world.[8]

Table 1: Annual incidence of IK per 100,000 population across various countries of the world.

Country/Region	Incidence per 100,000/ year
Nepal	799
Myanmar (Burma)	710
Bhutan	339
India (Madurai, Tamil, Nadu)	113
United Kingdom	40.3
United States	27.6
Australia	6.6
China (Prevalence Study)	192

AGE

Differences in incidences across age groups has also been noted with IK being commoner in ages between 30-55 years age group[9] which is attributed to contact lens wear and work-related trauma.[10] Elderly patients affected by IK also have poor outcome and higher complications such as perforations and loss of the eye[11] which was attributed to higher rate of ocular comorbidities and delay in presentation to hospital due to dependence on others for seeking medical care.[12]

SOCIO-ECONOMIC STATUS

Socio-economic status also affects the incidences of IK.[13] Low socio-economic status increases risk of developing IK due to poor education, lack of ocular protection, poor personal hygiene and limited access to eye care in rural areas.[14] Application of breast milk, plant and animal products into the eye are common practices in some countries of Africa which has been found to actually worsen IK.[15]

CAUSATIVE MICROORGANISMS

A wide range of organisms cause IK, these include:

Bacteria

Consist of Gram-positive and Gram-negative organisms as a result of the difference in composition of their cell wall. Gram positive bacteria have a thick outer cell wall, which is composed of layers of peptidoglycan interspersed with teichoic acids and lipoteichoic acids, whereas Gram negative bacteria consist of thin middle layer peptidoglycan and an additional outer membrane primarily made of lipopolysaccharide, which plays an important role in the pathogenesis of infection and to host inflammatory reaction.[16] Bacterial Keratitis is the most common type of IK across most regions of the world.[17] The organisms involved include; staphylococcus aureus, streptococcus spp, and Coagulase negative staphylococcus.[18] Other organisms include pseudomonas aeruginosa, enterobacteriaceae spp, corynebacterium spp and propionibacterium spp. Recently, over the past decade, Moraxella, Nocardia and Non-Tuberculous mycobacterium have been implicated in longer cornea healing time, steroid corneal ulcers and refractive surgery respectively.[19]

Fungi

Fungi can be filamentous and yeast or yeast-like. Filamentous fungi such as fusarium spp and aspergillus spp are implicated in IK across tropical regions while yeast and yeast like agents such as candida spp are more in temperate regions. Fusarium spp and aspergillus are main fungal causes of IK in Asia.[20] With improvements in diagnostic technology, pathogens such as Cryptococcus Curvatus, Arthrographis kalrae, pythium spp and others are being reported.[21]

Protozoa

Acanthamoeba, aubiquitous free-living protozoan is found in water, soil, air and dust.[22] Acanthamoeba Keratitis (AK) is associated with prolonged treatment course and poor visual outcome[23] and affects 1-33 per million contact lens wearers per year. Recent studies revealed AK was also detected in non-contact lens (CL) wearers following trauma or exposure to contaminated water, soil or dust[23] as it adheres to the hydrophilic plastics used in contact lenses. Microsporidia Keratitis is another type of parasitic IK and accounts for

0.4% of all cases of IK observed mainly in Asia.[24] It manifests as superficial kerato-conjunctivitis or stromal keratitis and is associated with exposure to contaminated water or soil and acquired immune deficiency syndrome.[24]

Viruses

Herpes Simplex Keratitis (HSK) ad Herpes Zoster Keratitis (HZK) are common causes of IK and represents an important cause of IK accounting for 15-21% in some regions.[25] Herpes keratitis often causes neurotropic keratopathy leading to poor corneal healing with increased risk to further IK with other corneal complications like melting and perforation.[25] Primary Herpes Simplex Virus (HSV) infection occurs in children and young adults as conjunctivitis while its ocular infection occurs in adults due to reactivation of HSV from primary infection. Herpes Zoster Virus (HZV) infection affects specific areas of the body (dermatomes) with 50 – 70% of HZV infection involving the eyes and presents as herpes zooster ophthalmicus.

POLY-MICROBIAL INFECTION

This form of IK is caused by two or more micro-organisms and has been reported in 2-15% of all IK cases.[26] These organisms may be from the same category e.g. bacteria-bacteria, fungus-fungus or different categories such as bacteria-fungus. This possesses a significant diagnostic and therapeutic challenge with worse outcome than mono-microbial.[26] Due to common occurrence of poly-microbial keratitis and low culture yield of microbial investigation, it is advisable to maintain a low threshold of repeating corneal scrapping where there is poor response to antibacterial or antifungal agents.

SEASONAL VARIATIONS

Pathogens adapt to climate and seasons and as a result, IK is commoner during the summer with pneumonia aeruginosa being one of the frequent isolates, this has been attributed to increase CL wear and engagement in water activities such as swimming.[27] Fungal keratitis cases are high during windy and harvest periods due to increased risk of trauma from agricultural activities.[28]

RISK FACTORS FOR IK

Contact Lens (CL) Wear

Contact Lens (CL) Wear has been shown to be one of the commonest risk factors in developed countries and found to be 9.3 times responsible for IK in CL wearers than non-wearers. CL wear is believed to decrease tear film exchange during blink thereby causing stagnation of tear under the CL which leads to accumulation of microbes and adherence to the cornea reducing corneal epithelial desquamation and alteration of tear fluid biochemistry. Multiple predisposing factors also come into play such as poor CL, CL case hygiene, overnight wear etc.[29] CL related keratitis is most commonly associated with pseudomonas aeruginosa and acanthamoeba spp and a major risk factor for fungal keratitis.[29]

Trauma

Trauma is also a common risk factor and studies have reported farmers and manual laborers to constitute occupation at risk of IK as a result of exposure to plant materials and foreign bodies which is worsened by lack of eye protection.[30]

Fungal keratitis is the commonest cause of trauma related IK in developed countries[30] while trauma related IK were caused more by Gram positive bacteria.[31]

Ocular Surface and Eyelid Diseases (OSDs)

These diseases include Steven-Johnson Syndrome, ocular cicatricial pemphigoid, bullous keratopathy, blepharitis just to mention a few. IK related to OSDs are caused mainly by Gram positive bacteria such as staphylococcus aureus.[30] The abnormal ocular surface can lead to breakdown of corneal epithelium which is important for defense leading to increased risk of IK from invading organisms.[30]

Post Ocular Surgery

Ocular surgeries such as cataract surgery, corneal transplant, refractive surgery and others may serve as opening for bacteria or other microorganisms to gain entry into the eye.[32]

With increase in number of refractive corneal surgery world-wide, IK from such procedures is becoming an important entity with most cases attributed to Gram positive bacteria and non- tuberculous mycobacterium.[32] though fungal and acanthamoeba infection may also occur. Corneal crosslinking (CXL) is another procedure which has been reported to increase risk of IK through reactivation of herpes keratitis.[33]

USE OF TOPICAL STEROIDS

Topical steroid is used in ophthalmology as anti-inflammatory/immune-modulatory agent. It has found use in post operative care after intraocular and ocular surface surgeries such as cataract surgery, corneal transplant, glaucoma surgery etc.[34] Studies have implicated topical steroids in fungal keratitis and poly-microbial keratitis.[34] While topical steroids are useful in treating stromal herpes simplex keratitis, it can exacerbate epithelial HSK leading to geographic ulcer and studies have linked AK with topical steroids.[34]

SYSTEMIC IMMUNOSUPPRESSION

Diabetes is associated with decrease corneal sensation, wound healing and neuropathic keratopathy[35] which increases the risk of IK from fungal, bacterial, viral and mixed organisms.[36]

ANTI-MICROBIAL RESISTANCE (AMR)

AMR can be developed by mutational resistance or horizontal gene transfer and has become a serious issue in the management of IK. This may be as a result of drug abuse, uncertainty in diagnosis leading to wrong prescription, financial incentives for prescribing antibiotics etc.[37] Broad spectrum topical antibiotic therapy is the sole standard in IK treatment involving use of aminoglycosides and cephalosporins or monotherapy with fluroquinolone.[38] High concentration of antibiotics is achieved at the target site during topical intensive antibiotic application thereby reducing emergence of resistance.

Recent cases of AMR in ocular infection had been reported in various parts of the world with various reasons given as the cause[39] and resistance varies with organisms and between drugs.[40]

DIAGNOSIS OF INFECTIOUS KERATITIS

Detailed history with presenting clinical feature may be sufficient to identify causative agents, however, a thorough investigation may also be required. Gram stain is useful for identifying bacterial and fungal agents. It is able to identify 60-75% of bacterial and 35-90% of fungal agents.[41] Culture on Blood Agar (BA) and Chocolate Agar (CA) is useful for isolating most bacteria agents while Sabouraud dextrose Agar (SDA) is effective for isolating fungal agents. Advanced diagnostic procedures such as Polymerase Chain Reaction (PCR) can give high sensitivity of isolates.[41]

In-vivo Confocal Microscopy (IVCM) has high sensitivity and rapidity in detecting filamentous fungi, acanthamoeba and nocardia bacteria.[42] Anterior Segment Optical Coherence Tomography has been successfully used recently to provide measure of corneal infiltrates and/or scar size and also monitor corneal thinning.[43]

TREATMENT OF INFECTIOUS KERATITIS

The best practice to avoid antibiotic resistance is to administer dual therapy using cephalosporin and aminoglycoside or a Monotherapy with fluoroquinolone. Intensive topical antibiotics applied directly and regularly increases concentration of drugs at site thus, enhancing recovery of IK and reducing development of resistance.[44] Gatifloxacin, a fourth-generation fluoroquinolone has been found effective against most gram-negative bacteria e.g. pseudomonasaeruginosa and acinetobacter spp and therefore commonly used as a monotherapy in Gram-negative IK.[45]

NEW TRENDS IN TREATING INFECTIOUS KERATITIS

NON-PHARMACEUTICAL

Ultraviolet Radiation is used to sterilize ocular surface before treating the disease with antibiotics, antifungal or as may be required. Loading antibiotics as nano-particles to enhance penetration into oculartissues especially for fluoroquinolones and amphotericin B to allow them reach higher concentrations is another method. Drug-eluting contact lenses are also being considered to overcome compliance while other approaches include micro-needles imprinted on the cornea surface to deliver drugs deep into the cornea tissue and light base therapies such as cornea CXL therapy are also being explored.[46]

PHARMACEUTICALS

There are new pharmaceuticals being tried in the treatment of IK and this include anti-microbial peptides (AMB) e.g. thymosin.

Cifiderocol, a systemic antibiotic is also showing promising antibacterial effect against pseudomonas in rabbit model of keratitis.combination of antibiotics e.g. polymyxin B/trimethoprim with rifampicin has improved the potency of the drug making the combination a broad-spectrum agent more potent than Moxifloxacin alone. With these novel modalities, the future for treatment of IK looks brighter than ever.[47]

CONCLUSION

IK highlights the constant burden on human health across the whole world. This burden may be underestimated therefore well-designed studies to include all microorganisms are required for the true impact of IK to be confirmed.

The role of major risk factors is important in order to enhance public health intervention so as to reduce IK risk. There is a need for judicious use of antimicrobials and development of newer potent agents along with strategies for therapies. Development of newer and improved technologies such as next-generation sequencing and artificial intelligence (AI) assisted platforms could provide better guidance on correct use of antimicrobial therapy in the future thereby reducing the risk of AMR.[48]

REFERENCES

1. Khurana A. Comprehensive Ophthalmology 5th Edition 2014 page 91. ISBN:978-81-224-3331-9. New Age International Publishers.
2. Cabrera-Aguas M, Khoo P, Watson LS. Infectious keratitis: A review. Clin Exp Ophthalmol 2022;50(5):543-62. Doi:10.1111/ceo.14113.
3. Ung L, Acharya NR, Agarwal T, Alfonso EC, Bagga B, Bispo PJM, et al. Infectious corneal ulceration: a proposal for neglected tropical disease status. Bull World Health Organ. 2019; 97(12):854-6.
4. Jeng BH, Gritz DC, Kumar AB, Holsclaw DS, Porco TC, Smith SD et al. Epidemiology of Ulcerative Keratitis in Northern California. Arch Ophthalmol 2010; 128:1022-8
5. Ibrahim YW, Boase DL, Cree IA. Incidence of infectious corneal ulcers, Portsmouth study, UK. J Clin Exp Ophthalmol 2012; S6 :001 [Google Scholar].
6. Gonzales CA, Srinivasan M, Whitcher JP, Smolin G. Incidence of Corneal ulceration in Madurai district, south India. Ophthalmic Epidemiol 1996; 3: 159-66.
7. Upadhyay MP, Karmacharya PC, Koirala S, Shah DN, Shakya S, Shresth JK et al. The Bhaktapureye study: Ocular trauma and antibiotic prophylaxis for the prevention of corneal ulceration in Nepal. Br J Ophthalmol , 2001; 85: 388-92. [Pub Med] [Google scholar] [PubMed central].

8. Lawson Ung MD, Paulo J.M. Bispo PhD, Swapna S. Shanbhag MD, Michael S, et al; the persistent dilemma of Microbial Keratitis: Global Burden, Diagnosis, and Antimicrobial Resistance Survey of Ophthalmol; 2018; 64(3): 255-271. doi 10.1016/i.survophthal.2018.12.003
9. Green M, Carnt N, Apel A, Stapleton F. Queensland Microbial Keratitis Database 2005-2015. Br J Ophthalmol 2019;103: 1481-6.
10. Keay LJ, Gower EW, Iovieno A, Oechsler RA, Alfonso EC, Matoba A et al. Clinical and Microbiological Characteristics of Fungal Keratitis in the US 2001-2007: a multicenter study. Ophthalmology2o11; 118: 920-6. [Google scholar].
11. Butler TK, Spenser NA, Chan CC, Singh GJ, McClelan K. Infective keratits in older patients: a 4-year review, 1998-2002. Br J Ophthalmol 2005; 89: 591-6
12. Kunimoto DY, Sharma S, Garg P, Gopinathan U, Miller D, Rao GN. Corneal Ulceration in the elderly in Hyderabad, south India.Br J Ophthalmol. 2000; 84:54-9. doi:10.1136/bjo.84.154. [DOI] [PMC free article] [Pub Med] [Google scholar].
13. Mandour SS, Marley HM, Farahat HG. Resistant Microbial Keratitis in South Nile Delta, Egypt: Influence of Regional Risk Factors. Semin Ophthalmol2016; 31: 437-8. Doi: 10.3109/0882-0538.2014. 962174 [DOI] [Pub Med] [Google scholar]
14. Gautam V, Chaudhary A, Singh SK, Rai PG. Profile of Corneal ulcer in a month of harvesting season in a tertiary Eye Hospital of Eastern Nepal. Nepal J Ophthalmol 2018; 10:32-8. doi: 10.3126/nepjoph. v10i1.216686. [DOI] [Pub Med] [Google scholar].
15. Oladigbolu K, Rafindadi A, Abah E, Samaila E. Corneal ulcers in a tertiary hospital in Northern Nigeria. Ann Afr Med. 2013; 12:165-70. doi:10.4103/1596-3519.117626 [DOI] [Pub Med] [Google scholar].
16. Silhavy TJ, Kahne D, Walker S. The bacterial cell envelope. Cold Spring Harb Perspect Biol. 2010; 2: a000414. [DOI] [PMC free article] [Pub Med] [Google scholar].
17. Tam ALC, Cote' E, Saladanha M, Lichtinger A, Slomovic AR. Bcterial Keratitis in Toronto; a 16-year Review of Microorganisms isolated and the Resistance Patterns Observed. Cornea. 2017;36: 1528-34. doi: 10.1097/ICO.0000000000001390. [DOI] [Pub Med] [Google scholar].
18. Peng MY, Cevallos V, Mcleod SD, Lietman TM, Rose Nussbaumer J. Bacteria Keratitis: Isolated Organisms and antibiotic resistance patterns in San Francisco. Cornea 2018;37: 84-7.doi:10.1097/ ICO.0000000000001417. [DOI] PMC free article] [Pub Med] [Google scholar].
19. Kaliamurthy J, Kalavathy CM, Parmar P, Nelson-Jesudasan CA, Thomas PA. Spectrum of bacterial keratitis at a tertiary eye center in India. Biomed Rest Int. 2013; 2013: 181564. Doi: 10.1155/2013/181564 [DOI] [PMC free article] [Pub Med] [Google scholar].
20. Lalitha P, Prajna NV, Manoharan G, Srinivasan M, Mascarenhas J, Das M, et al. Trends in bacterial and fungal keratitis in south India, 2002-2012. Br J Ophthalmol 2015; 99:192-4. doi: 10.1136/bjophthalmol 2014-305000 [DOI] [PMC free article] [Pub Med] [Google scholar].
21. Sahay P, Goel S, Nagpal R, Maherana PK, Sinha R, Agarwal T et al. Infectious keratitis caused by Rare and Emerging Microorganisms. Curr Eye Res 2020; 45: 761-73. doi: 10 .1080/ 02713683.2019.1708407. [DOI] [Pub Med] [Google scholar].

22. Somani SN, Ronquillo Y, Moshirfar M. Acanthamoeba Keratitis. Stat pearls. Treasure Island (FL): stat pearls publishing LLC; 2020. [Pub Med] [Google scholar].
23. Garg P, Kalra P, Joseph J, Non-contact lens related canthamoeba Keratitis. Indian J Ophthalmol 2017; 65: 1079-86. doi: 10.4103/ijo.IJO_826_17. [DOI] PMC free article] [Pub Med] [Google scholar].
24. Friedberg DN, Stenson SM, Orenstein JM, Tierno PM, Charles NC. Microsporidial keratoconjunctivitis in acquired immunodeficiency syndrome. Arch.Ophthalmol.1990;108:504-8. doi:10.1001/archopt.1990.01070060052047. [DOI] [Pub Med] [Google scholar].
25. Tuli S, Gray M, Shah A. Surgical Management of herpetic keratitis. Curr Opin Othalmol 2018; 29:347-54. doi: 10.1097/ICU.0000000000000484. [DOI] [Pub Med] [Google scholar].
26. Lim NC, Lim DK, Ray M. Polymicrobial versus Monomicrobial keratitis: a retrospective Comparative study. Eye Contact lens. 2013; 39:348-54. doi:10.1097/ICL.0b013e318a3024e. [DOI] [Pub Med] [Google scholar].
27. Khor HG, Cho I, Lee K, Cheng LL. Spectrum of microbial keratitis Encounterd in the tropics. Eye Contact Lens 2020; 46:17-23. doi: 10.1097/ICL.0000000000000621. [DOI] [Pub Med] [Google scholar].
28. Kowalski RP, Nayar SV, Romanowski EG, Shanks RMQ, Mammen A, Dhaliwal DK et al. The prevalence of Bacteria, Fungi, Viruses and Acanthamoeba From 3,004 cases of keratitis, Endophthalmitis and Conjunctivitis. Eye Contact Lens. 2019; 10.1097/ICL.0000000000000642. [DOI] [Pub Med].
29. Fleiszig SMJ, Kroken AR, Nieto V, Grosses MR, Wan SJ, Metruccio MME, et al. Contact Lens-related corneal infection: Intrinsic resistance and its compromise. Prog Retin Eye Res2020; 76:100804. doi: 10.1016/j.preteyeres. 2019.100804. [DOI] [PMC free article] [Pub Med] [Google scholar]
30. Pan XJ, Jiang T, Zhu H, Liu PP, Zhou ZY, Mao AJ. Corneal infection in Shandong peninsula of China: a 10-year retrospective Study on 578 cases. Int J Ophthalmol 2016; 9:53-7 doi:10.1824/ijo.2016.01.09 [DOI] [PMC free article] [Pub Med] [Google scholar].
31. Khoo P, Cabrera-Aguas M,Robaei D, Lahra MAA, Watson S. Microbial keratitis and Ocular surface disease: a 5-year study of the microbiology, Risk factors and clinical outcomes in Sydney Australia. Curr Eye Res 2019; 44:1195-202. doi:10.1080/02713633.2019.1631852. [DOI] [Pub Med] [Google scholar].
32. Di Zazzo A, Kheirkhah A, Abud TB, Goyal S, Dana R. Management of high-risk Corneal transplantation. SUN Ophthalmol. 2017; 62:816-27 doi: 10.1016/j.survophthal.2016.12.010 [DOI] [PMC free article] [Pub Med] [Google scholar].
33. Maharana PK, Sahay P, Suyeth M, Singhal D, Rathi A, Titiyal JS et al. Microbial keratitis After Accelerated Corneal Collagen Cross-linking in Keratoconus. Cornea 2018; 37:162-7. doi: 10.1097/ICO.0000000001437. [DOI] [Pub Med] [Google scholar].
34. Wilhemus KR, Gee L, Hulek WW, Kurinij N, Dawson CR, Jones DB et al. Herpetic Eye Disease Study: a controlled Trial of Topical Corticosteroids for Herpes Simplex Stromal Keratitis. Ophthalmology 2020.01.037 [DOI] [Pub Med] [Google scholar].

35. Al-Agba MA, Dhilon VK, Mohammed I, Said DG, Dua HS. Corneal Nerves in Health and disease. Prog Retin Eye Res 2019; 73:100762. DoI: 10.1016/j.preteyeres. 2019.05.003. [DOI] [Pub Med] [Google Scholar].
36. Kaiserman I, Kaiserman N, Nakar S, Vinker S. Herpetic Eye Disease in Diabetic Patients. Ophthalmology 2005.070.014 [DOI] [PubMed] [Google Scholar]
37. Munita JM, Arias CA. Mechanism of Antibiotic Resistance. Microbiol Spectr. 2016;4. 10.1128/microbiol spec. VMBF-0016-2015. [DOI] [PMC free article] [Pub Med].
38. McDonald EM, Ram FS, Patel DV, McGhee CN. Topical antibiotics for the management of bacterial keratitis: an evidence-based review of high quality randomized cotrolled trials. Br J Ophthalmol 2013-304660 [DOI] [Pub Med] [Google scholar].
39. Brown L. Resistance to ocular antibiotics; an overview. Clin Exp Optom 2007; 90:258-62. doi: 10.1111/j.1444-0938. 20007.00154.x. [DOI] [Pub Med] [Google scholar].
40. Bloomquist PH, Methicilin-Resistance Staphylococus aureus infections of the eye and orbit (an American Ophthalmological Society Thesis) Trans Am Ophthalmol Soc. 2006; 104:322-45. [PMC free aricle] [Google Scholar].
41. Badiee P, Nejabat M, Alborzi A, Keshavarz F, Shakiba E. Comparative study of Gram stain, potassium hydroxide smear, culture and nested PCR in the diagnosis of fungal keratitis. Ophthalmic research. 2010;44(4): 251-256 doi:10.1159/000313988 [DOI] [Pub Med] [Google scholar].
42. Nielsen E, Heegaard S, Prause JU, IvarsenA, Mortensen KL, Hjortdal J. Fungal keratitis -Improving diagnostics by confocal microscopy. Case reports in ophthalmology. 2013 sep;4(3):303-310. doi: 10.1159/000357558 [DOI] [PMC free article] [Google scholar].
43. Konstantopoulos A, Kuo J, Anderson D, Hossain P. Assessment of the use of Optical Coherence Tomography in microbial keratitis. American journal of ophthalmology. 2008 oz; 146:534-542. doi: 10.1016/j.ajo. 2008.05.030. [DOI] [Pub Med] [Google scholar].
44. Sneha S, Manish R, Sumeet K, Dhull CS, Sumit S, Jitender P. Recent trends; Medical management of infectious keratitis. Oman J Ophthalmol 2015; 8(2):83-85. doi:10.4103/0974-620x.159104.
45. Hanet MS, Jamart J, Chavas AP, Flouroquinolones or Fortified Antibiotics for treatment of Bacteria Keratitis: Systemic Review and Meta-analysis of comparative studies. Can J Ophthalmol 2012; 47:493-9 doi: 10.1016/j.jcjo.2012.09.001. [DOI] [PubMed] [Google Scholar].
46 Racheal W. Lecture: Infectious keratitis: Management and Emerging Therapeutics. https:// cybersight.org
47. Ung L, Bispo PJM, Doan T, Van Geldes RN, Gilmore MS, Lietman T et al. Clinical Metagenomics for infectious corneal ulcers: Rags to Riches? Ocul Surf 2020; 18: 1-12. doi: 10.1016/j.jtos 2019. 10.1 10.007 [DOI] [PMC free article].
48. Ting DSJ, Foo VH, Yang LWY, Sia JT, Ang M, Lin H et al. Artificial Intelligence for anterior segment diseases: emerging applications in ophthalmology. Br J Ophthalmol 2020; 10.1136/bjophthalmol.2019-315651 [DOI] [PubMed].